

Syllabus: Digital Logic Design

Course No.	Title of the Course	Course Structure	Prerequisite
	Digital Logic Design	3L-0T-2P	–
COURSE OUTCOMES (COs) CO1: To get familiarized with number systems, codes, logic gates, logic families and Boolean algebra CO2: To analyze and understand the design process associated with combinational Circuits. CO3: To analyze and understand the design process associated with sequential circuit. CO4: To understand fundamental concepts of VHDL modeling for basic digital circuits. CO5: To develop basic understanding of programmable logic devices and convertors			
UNIT	CONTENTS		
1	Introduction to Digital Systems: Number Systems: Binary, octal and hexadecimal number systems, Number-Base Conversions, Complements of Numbers. Binary addition and subtraction (unsigned and one's and two's complement), Different codes and their properties (e.g., Gray, BCD, 84-2-1, 2421 etc.), Boolean equations and terminology (SOP, POS, minterm, maxterm), Logic implementations of equations (2-level, AND/OR, NAND/NOR, etc.), Boolean algebra axioms		
2	Karnaugh Maps up to 5-variables with don't care conditions, Tabulation method, Parallel adders, BCD adder, Multiplexers, Multiplexers, Encoders, Decoders, Priority encoders, Comparators, Code converters		
3	Latch versus flip-flop, SR and D latch, D, SR, JK, and T flip-flops, Registers, Counters, Shift registers, Design & Analysis, Problem description to state table/graph, State assignment (binary, Gray), Next state and output equations, Mealy & Moore models of finite state machine, sequence detector and other examples.		
4	Hardware Language(VHDL) Introduction and its types of modeling style, VHDL implementation of adder, subtractor, multiplexer, decoders, comparators etc, VHDL Implementation of Flip Flops, latches, Registers, Counters		

5	Properties and characteristics (logic thresholds, delay, Noise margin, fan-in and fan-out, power dissipation) of logic families, Transistor-level schematics of CMOS logic gates, Introduction to Digital to analog convertor, Analog to digital convertor, Introduction to Programmable Logic Devices: PROM, PLA, SRAM, DRAM
SUGGESTED READINGS: 1. Digital Design and Computer Architecture. David Harris and Sarah Harris (Morgan Kaufman). 2. Digital principles & design : Donald D. Givone, McGraw Hill 3. Digital logic and computer design : M. M. Mano, PHI 4. Charles Roth," Digital Systems Design using VHDL", 5th Edition Cengage Learning 5. R.P. Jain, 'Modern Digital Electronics', McGraw Hill education, 4th Edition	

List of Experiments

- (1) Verify the truth table of AND, OR, NOT, NAND, NOR, X-OR, X-NOR gates
- (2) Implement all the above mentioned gates by using NAND gates and NOR gates only.
- (3) Design and Implement Half-adder, Full-adder, Half-subtractor, Full-subtractor using logic gates.
- (4) Design a 4 bit parallel adder and subtractor using IC. Further using the same IC implement BCD to excess-3 code convertor.
- (5) Design a 4 bit magnitude comparator using IC. Also implement 2 bit magnitude comparator using gates only.
- (6) Design and implement a full adder circuit using DECODER and gates. Also implement the same by using complimentary output decoder.
- (7) Design the following Flip-flop using NAND/NOR gates
 - (i) S-R FF (ii) D FF (iii) J-K FF (iv) T FF
- (8) Design and implement a MOD 6 synchronous UP counter using T FF.
- (9) Design a 2 bit UP/DOWN counter using J-K FF

Implement the following circuits using VHDL

- (10) Full adder and subtractor
- (11) 4X1 MUX and 3X8 decoder
- (12) Conversion of BCD to Excess-3 code

Database management system

COURSE CONTENT

UNIT 1

Introduction: Database management system Characteristics of the Database, Database Systems and Architecture, Data Models, Schemes & Instances, DBMS Architecture & Data Independence, Database administrator & Database Users, Database Languages & Interfaces, DDL, DML, DCL, Overview Relational Data Base Management Systems

UNIT 2

Data Modeling: Data modeling using The Entity-Relationship Model – Entities, Attributes and Relationships, Cardinality of Relationships, Strong and Weak Entity Sets, Generalization, Specialization, and Aggregation, Translating your ER Model into Relational Model, Relationships of higher degree.

UNIT 3

Relational Model, Languages & Systems: Relational Data Model concepts, Relational Model Constraints, integrity constraints, Keys domain constraints, referential integrity, assertions triggers, foreign key
Relational Algebra and calculus, SQL. Database security.

Relational Data Base Design: Functional Dependencies & Normalization for Relational Databases, Functional Dependencies, Normal Forms Based on Primary Keys, (1NF, 2NF, 3NF & BCNF), Lossless Join and Dependency Preserving Decomposition, Functional dependencies and its closure, covers and equivalence.

UNIT 4

Transaction Management: Transaction Concept and State, Implementation of Atomicity and Durability, Concurrent Executions, Serializability: Testing of serializability, Serializability of schedules, conflict & view serializable schedule.

Concurrency Control Techniques: Lock-Based Protocols, Timestamp-based Protocols, validation based protocol. Deadlock Handling

UNIT 5

Recovery System

Recoverability: Failure Classification, Storage Structure, Recovery and Atomicity, Log-based Recovery, Shadow Paging, Recovery with Concurrent Transactions

Storage organization : Indexing, Hashing ,file storage.

List of Experiments:

Course No.	Title of the Course	Course Structure	Pre-Requisite
CMMTC13	Applied Linear Algebra	3L-1T-0P	None

COURSE OUTCOMES (CO)

1. . To understand the detailed concept of Modern Algebraic.
2. To impart the knowledge of Vector Space.
3. To know about Linear Transformations.
4. To know the concept of bilinear form and Inner Product of vector spaces.

COURSE CONTENT:

Introduction to Modern Algebra: groups, ring and field, Vector spaces, properties of vector spaces, subspaces, linear dependence and independence, Linear span, Bases and dimension, direct sums.

Linear transformations: Linear transformations, Range and null spaces, rank –nullity theorem and its application, projections, Inverse linear transformation, representation of linear transformations by matrices, change of basis, Dual space, Dual bases,

Trace and determinant of linear transformation, Transpose of a linear transformation, Eigen values and eigen vectors, Minimal polynomials, Cayley–Hamilton theorem, diagonalization, Jordan canonical form, matrix norms.

Inner product spaces: Inner product spaces, norm of a vector, orthogonality, orthonormal set, orthonormal basis, Gram-Schmidt orthonormalization, orthogonal projections, Linear functional and adjoints, Hermitian, self-adjoint, Unitary and normal operators.

Bilinear forms: Bilinear forms, symmetric and skew-symmetric bilinear forms, real quadratic forms, positive definiteness.

Recommended Books:

1. K. Hoffman and R. Kunze, Linear Algebra, 2nd edition, Pearson education, 2015
2. G. Hadley, Linear Algebra, Narosa, 2002
3. Gilbert Strang, Linear Algebra and its applications, Cengage Learning, 4th edition, 2007
4. Serge Lang, Introduction to Linear Algebra, Springer, 2004
5. Linear Algebra, Fourth edition, Schaum's outlines, 2015

Probability & Statistics (Iguen)

COURSE CONTENT:

Important Theoretical Distributions: Review of continuous and discrete probability distributions, Poisson process, Negative binomial distribution, Fitting of standard distributions, Fitting of Normal distribution by method of areas and method of ordinates, Hypergeometric distributions, Multinomial distribution and their properties.

Continuous Frequency Distribution: Moments, Sheppard's corrections, Kurtosis, Coefficient of variation, Rectangular distribution, Beta distribution of first and second kind, Gamma distribution, Cauchy's distribution, Geometrical probability, Tchebycheff's and Markov's inequalities.

MGF and Method of Least Square: Change of origin and scale in mgf, moment generating functions of standard distributions (Poisson, Binomial, Exponential, Uniform, Normal, Gamma, chi square), Cumulants, characteristic function, Weak law of large numbers, Central limit theorem. Method of least squares: Fitting of straight lines, parabola and exponential curves.

Simple sampling of attributes: Large samples, mean and S. D. in simple sampling of attributes, Test of significance for large samples, Standard error, Type I and II errors, Null hypothesis, Confidence limits, Chi-square distribution, Degree of freedom, Level of significance, Test of goodness of fit, Test of independence, Coefficient of contingency, Yate's correction for continuity.

Sampling of variables: Small samples, t-distribution, test of significance of the mean of random sample from normal population, F-distribution, Relationship between t, F and chi square distributions, ANOVA (one-way criterion only).

Recommended Books:

1. **An Introduction to Probability Theory and Its Applications, Vol. 1 (Wiley Series in Probability and Statistics) by W. Feller**
2. **Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Publication)**
3. **Probability & Statistics- SOS by Spiegel, McGraw Hill**
4. **Probability and Statistics for Engineers by Anthony J. Hayter (Cengage Learning)**
5. **Mathematical Statistics, Krishna Prakashan Media by J. K. Goyal and J. N. Sharma**

No.	Title of the Course	Course Structure	Pre-Requisite
EC301/ ETEC301/ MMTC301/	Engineering Mathematics	3L-1T-0P	None

COURSE OUTCOMES (CO)

1. To know the concepts of the Fourier series, Harmonic analysis, and its applications
2. To know the concepts of vector calculus and its applications.
3. To know the concepts of Laplace Transforms and its application.
4. To understand the concepts of Legendre of the first kind with their properties
5. Study and apply the method of solution for special Partial Differential Equations

Unit 1: Fourier Series: Periodic functions, Fourier series, Functions of any period, Even and odd functions, Half range series, Harmonic analysis.

Unit 2: Vector Calculus: Differentiation of a vector function, scalar and vector fields, Gradient, Divergence, Curl, line integral, independence of path, Green's theorem and applications, Surface Integral, Stoke's theorem, and applications; Volume Integrals, Gauss Divergence theorem, and applications.

Unit 3: Laplace Transform: Basic properties, Laplace transform of derivatives and integrals, Laplace transform of periodic functions, Unit step function, Dirac-Delta function, change of scale property, Shifting theorems, Inverse Laplace transform, convolution theorem, applications of L. T. to solve ordinary differential equation.

Unit 4: Legendre function: Legendre's equation and its solution, Legendre function of the first kind, Generating function for Legendre polynomials, Recurrence relations, Rodrigue's formula, Orthogonal properties of Legendre's polynomials.

Unit 5: Partial Differential Equations: Introduction to PDEs and solutions of Heat and Wave equations (one dimensional only), Laplace equation and simple problems based on these.

Recommended Books:

1. Jain and Iyenger, "Advanced Engineering Mathematics," 5th edition, Narosa Publication, 2017.
2. E. Kreyszig, "Advanced Engineering Mathematics," 10th edition, Wiley Publication, 2020.
3. M. D. Greenberg, "Advanced Engineering Mathematics," 2nd edition, Pearson Publication, 1998.
4. A. S. Gupta "Calculus of Variations with Applications," PHI Publication, 2011.